Diem’s torus charts: disproof and repair

Source-helper failure. Original Diem2011 Proposition2.5(b), printedp81, uses a two-copy multiplicative-group cover of the projective line for every primepowerq. Lean disproves the rational-point consequence overF2 using actual Mathlib Projectivization(F2,F2×F2), not an assumed three-element model. Its cardinality is3; actual F2units have cardinality1. Any two maps fail to cover. All positive products also defeat2^ncharts. Signed standing unverified. [pcc:unverified]

Actual F2 obstruction: P1 has3 points; each torus has1 point.

[1:0]	[1:1]	[0:1]

Three injective charts: [1:u], [1:u−1], [u−1:1] (u≠0).

Products need3^n charts overF2. Proved overhead: 3^n ≤5n!.

Exact qualification. For every finite fieldK, two rational-point maps from K-units cover the actual projective line if and only if cardK≥3. Necessity follows from actual surjection/cardinality bounds. Sufficiency choosesa≠0,1 and proves that [1:u] and [u−1:u−a] are injective and cover. This includes fields of characteristic2 with more than2elements; characteristic2alone is not the obstruction. [pcc:unverified]

Uniform repair. Three explicit injective rational-point charts cover over every field: [1:u], [1:u−1], [u−1:1], withu nonzero. Actual projective induction and nonzero-scaling equivalence prove the cover, including both coordinate-axis points. Product covers hold at everyarity. OverF2, any cover fromm actual n-tori requires m≥3^n, even allowing arbitrary maps. [pcc:unverified]

Call-count allowance. The actual chart-index type has3^nelements, and Lean proves3^n≤5n! for every naturaln. For d>0 and B=n!d^n, it proves3^nB≤5B². Thus the repaired chart count has a polynomial allowance in the stated size parameter. This is arithmetic accounting, not a formal proof of Rojas’ runtime or the full decomposition algorithm. [pcc:unverified]

Primary-source check. C.Diem, Compositio Math147(2011),75–104, DOI10.1112/S0010437X10005075. Read notation/setup p78, Definition2.4 and Proposition2.5 pp80–81, its proof and laterCondition2.7; printedp81 visually inspected in full. Proposition2.5 explicitly acceptsq2; laterCondition2.7 is not its hypothesis. PDF878749B, SHA74902b58…a934a58a. Exact full corpus node THM-SP-DIEM11-DECOMPOSITION-ALG states the bound/main algorithm and is not refuted. No full-paper read or novelty claim. [pcc:unverified]

Checks and evidence. Whole gate 31 declarations /0 failures /0 trusted native exceptions; Lean4.35.0, Mathlib56a00585…. Closed declaration, all custom source/config and evidence/report hashes are pinned. All failed/successful source/log/exits and first no-sources gate failure are retained losslessly. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, new global flags or waivers. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Remaining scope. The literal source cover step fails. The main geometric Bezout bound and decomposition theorem are not refuted. Explicit rational-point formulas and chart arithmetic do not formalize scheme open embeddings, multihomogeneous substitution, isolated-point multiplicities, toric resultants or the full algorithm runtime. Existing solver-length/support evidence remains unchanged. KB unavailable; bounded source/formal fallback retained. No new native experiment, paid compute or agent upload. Other all-fourteen obligations remain open at their recorded scopes. [pcc:unverified]